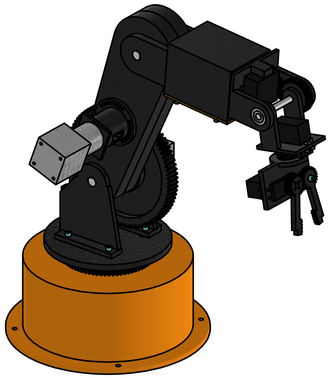
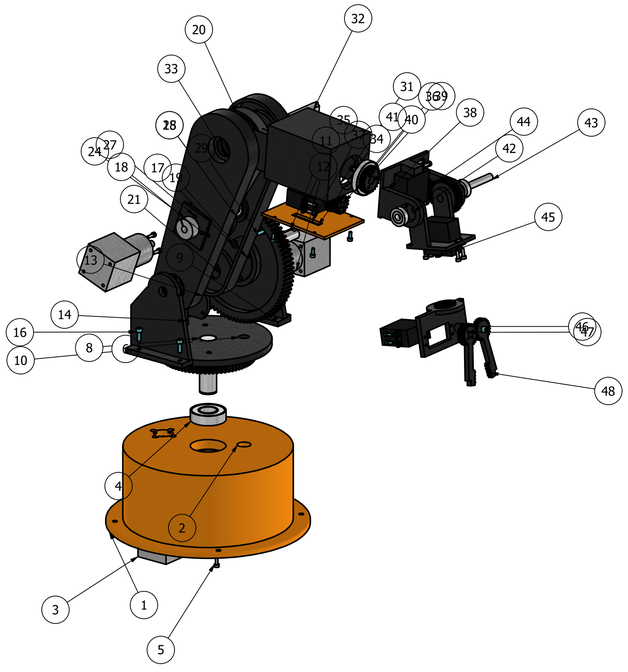
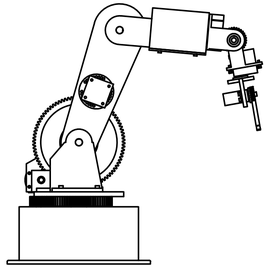
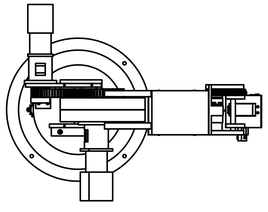








PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
25	1	HTD5M-15T-15-B8	Timing pulley HTD 5M 15T, 15 mm belt, 8 mm bore, aluminium
26	1	RA-213	J3 jackshaft dia 8 x 56.4, D-flat: carries the 15T pulley between two 608s
27	1	RA-117	J3 drive mount: eccentric spigot with the B-side 608 (turn it to tension the belt, +-1.5 mm), stand-off tube around the jaw coupling, gearbox plate with radial slots for a 28-30 mm bolt circle
28	1	RA-118	J3 eccentric bearing ring (A side): A-side 608 of the J3 jackshaft, turned with the drive mount to tension the belt, locked by an M3 set screw from the arm rim
29	4	ISO4762-M4x30	Socket head cap screw ISO 4762 M4x30, A2
30	4	ISO4762-M4x16	Socket head cap screw ISO 4762 M4x16, A2
31	1	RA-108	Forearm: elbow fork, J4 servo bay with ear posts, 6806 nose bearing seat
32	1	RA-210	Elbow shaft dia 12 x 64, D-flats for pulley and forearm fork
33	1	RA-209	J3 driven pulley HTD 5M 45T, 16 mm face, D-bore 12 (3:1 with vendor 15T)
34	1	RA-109	Forearm cover 3 mm (servo bay lid)
35	2	DS3235-270	Digital servo DSSERVO DS3235 (270 deg), 32 kg.cm @6 V stall, 40x20x40.5 case
36	1	6806-2RS	Deep-groove ball bearing 6806-2RS (30x42x7)
37	11	ISO4762-M3x8	Socket head cap screw ISO 4762 M3x8, A2
38	1	RA-110	Wrist yoke (J4 output): 6806 hub, J5 servo mount, 2x 608 seats for the J5 shaft
39	1	RA-208	J4 gear 24T m1.25 on the wrist-yoke hub (1.5:1 with the 16T servo pinion)
40	1	RA-206	J4 pinion 16T m1.25, bolted to 25T servo horn
41	4	HORN-25T-AL	Aluminium 25T round servo horn dia 25 (disc type)
42	1	RA-111	Wrist bracket (J5 output) with integral 32T m1.25 gear, J6 servo crossbar
43	1	RA-211	J5 shaft dia 8 x 84 (fixed in bracket, runs in 2x 608)
44	1	RA-207	J5 pinion 16T m1.25, bolted to 25T servo horn
45	2	DS3218-270	Digital servo DSSERVO DS3218 (270 deg), ~20 kg.cm @6 V stall, 40x20x40.5 case
46	1	RA-112	Gripper base: J6 horn flange, gripper servo mount, finger pivot
47	1	RA-113	Gripper finger L (servo-driven) with integral 24T m1 gear
48	1	RA-114	Gripper finger R (idler) with integral 24T m1 gear

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	RA-101	Base plate / bench flange, 4x M5 bench bolts on PCD 220
2	1	RA-102	Base housing: J1 bearing boss (2x 6004), J1 motor deck, cable exit
3	1	17HS19-1684S-PG5	Stepper StepperOnline 17HS19-1684S-PG5: NEMA17 48 mm, 1.68 A, 52 N.cm holding + PG5 planetary (5.18:1, 90%, 3 N.m max permissible)
4	2	6004-2RS	Deep-groove ball bearing 6004-2RS (20x42x12)
5	28	ISO4762-M3x10	Socket head cap screw ISO 4762 M3x10, A2
6	1	RA-103	Turntable with integral 72T m1.5 ring gear (J1), shoulder bracket mount
7	1	RA-204	J1 pinion 18T m1.5, D-bore 8 (7.5 across the flat) on the PG5 D-cut shaft, Loctite 638
8	1	RA-201	J1 shaft dia 20 x 62, h6, circlip groove (DIN 471-20)
9	1	RA-104	Shoulder cheek, motor side: outer 608 seat of the J2 jackshaft, stand-off mount, shaft clamp, bolted foot
10	1	RA-105	Shoulder cheek, idler side: shaft clamp, bolted foot
11	1	RA-202	Shoulder shaft dia 12 x 74 (stationary), flats for M4 set screws
12	2	17HS19-1684S-PG19	Stepper StepperOnline 17HS19-1684S-PG19: NEMA17 48 mm, 1.68 A, 52 N.cm holding + PG19 planetary (19.2:1, 80%, 3 N.m max permissible)
13	1	RA-116	J2 motor stand-off: flange on the cheek, tube around the jaw coupling (2 access windows), gearbox plate with radial slots for a 28-30 mm bolt circle
14	1	RA-115	J2 bearing block: inner 608 of the J2 jackshaft (pinion carried between two bearings), 2x M4 to the turntable
15	6	608-2RS	Deep-groove ball bearing 608-2RS (8x22x7)
16	5	ISO4762-M4x12	Socket head cap screw ISO 4762 M4x12, A2
17	1	RA-106	Upper arm half A (+Y): J2 gear side, 6201 seats, belt channel
18	1	RA-107	Upper arm half B (-Y): J3 drive-mount housing (eccentric belt tension), 3 M3 inserts, 6201 seats
19	1	RA-203	J2 driven gear 86T m1.75 (20 deg, face 14), bolted to upper arm A
20	1	HTD-375-5M-15	Timing belt HTD 5M, 375 mm closed loop (75 teeth), 15 mm wide
21	4	6201-2RS	Deep-groove ball bearing 6201-2RS (12x32x10)
22	1	RA-205	J2 pinion 16T m1.75, D-bore 8 on the J2 jackshaft flat, held axially between the two 608s
23	1	RA-212	J2 jackshaft dia 8 x 48.4, D-flat: carries the J2 pinion between two 608s
24	2	MJC25-8-A + JD16/25-98R	Jaw coupling Ruland: 2x MJC25-8-A clamp hubs (8 mm) + JD16/25-98R spider; 25.4 OD x 35.3, 8.48 N.m rated / 17 N.m peak

ROBOTICARM6DOF - 6-DOF ARM, 0.5 KG PAYLOAD
REACH 538 MM (SHOULDER-TCP, HORIZONTAL); L1/L2/L3/D6 = 204/200/192/146
MOVING MASS 2.09 KG, TOTAL 5.26 KG (INVENTOR)
J1 PG5 + 18:72 (20.72:1) J2 PG19 + 18:84 (103.2:1) J3 PG19 + HTD 15:45 (57.6:1)
J4 DS3235 + 16:24 M1.25 J5 DS3235 + 16:32 M1.25 J6 DS3218 DIRECT GRIPPER DS3218 + 24:24
J2 REQ 6.65 N.M / CAP 15.32 N.M (SF 2.3)
JOINT ANGLES = ASSEMBLY PARAMETERS q1..q6, grip (FULLY CONSTRAINED, 0 DOF)
EXPLODED VIEW = POSITIONAL REPRESENTATION 'Exploded'. FASTENERS ISO 4762 A2, THREADS COSMETIC

DRAWN	Jay Patel	11/1/2026	
CHECKED			
QA			TITLE: RoboticArm6DOF - general assembly
MFG			
APPROVED			
		SIZE: D	DWG NO: RA-DWG-001
		SCALE: 0.29 : 1	SHEET 1 OF 1